

ADEOLU ISRAEL MESOGBORIWON

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RESEARCH INTERESTS

- Industrial Artificial Intelligence (Industrial AI)
- Process Systems Engineering
- Smart Manufacturing and Industry 4.0
- Process Monitoring and Fault Diagnosis
- Predictive Maintenance
- Digital Twin Systems
- Industrial Automation and Control
- Machine Learning for Industrial Processes
- Sustainable Manufacturing Systems
- Industrial Internet of Things (IIoT)

EDUCATION

Diploma in Industrial Automation / Mechatronics Engineering

Institute of Industrial Technology (IIT), Lagos, Nigeria

March 2026

Relevant Areas:

- PLC Programming
- HMI/SCADA Systems
- Industrial Instrumentation
- Motor Control Systems
- Electro-Pneumatic Systems
- Industrial Communication Networks

Bachelor of Engineering (B.Eng.) Chemical Engineering

Federal University of Petroleum Resources, Effurun (FUPRE), Nigeria

November 2019

Class: **Second Class Upper Division (2:1)**

Relevant Coursework:

- Process Control
- Chemical Process Design

- Process Dynamics
- Heat and Mass Transfer
- Fluid Mechanics
- Numerical Methods
- Statistics
- MATLAB Programming

RESEARCH PROJECTS

Preliminary Design of a 3-Ton Chlorobenzene Production Plant

Final Year Design Project

- Designed a complete chlorobenzene production facility.
- Developed process flow diagrams and material balances.
- Performed equipment sizing and economic analysis.
- Optimized plant operation and production efficiency.

Analysis of the Impact of Crude Oil Pollution on Soil Physicochemical Properties Using Response Surface Methodology

Final Year Research Project

- Investigated the effects of crude oil contamination on soil properties.
- Applied Design Expert software and Response Surface Methodology (RSM).
- Developed predictive models and optimization strategies.
- Conducted statistical analysis and model validation.

INDUSTRIAL AUTOMATION PROJECTS

PLC-Based Tank Level Control System

- Developed automated tank level monitoring and control systems.
- Integrated sensors, PLC logic and HMI visualization.
- Implemented safety interlocks and fault handling mechanisms.

Smart Conveyor Sorting System

- Designed PLC-controlled automated sorting processes.
- Integrated sensors and pneumatic actuators.
- Improved sorting efficiency and reduced manual intervention.

Industrial SCADA Monitoring System

- Developed real-time process visualization systems.
- Implemented alarm management and historical data logging.

- Improved process monitoring and operational visibility.

DATA SCIENCE AND ANALYTICS PROJECTS

COVID-19 Data Analytics and Predictive Modeling

- Performed exploratory data analysis on global datasets.
- Built machine learning classification models.
- Conducted feature engineering and model evaluation.

Tools: Python, NumPy, Pandas, Scikit-Learn, Matplotlib & Jupiter Notebook

Sales Performance Dashboard

- Developed Power BI dashboards.
- Analyzed revenue trends and customer segments.
- Generated business insights for decision-making.

Tools: Jupiter Notebook & Power BI

PROFESSIONAL EXPERIENCES

Industrial Automation Engineer Trainee

Institute of Industrial Technology (IIT), Lagos, Nigeria

August 2025 – March 2026

Areas of Interest:

- PLC Programming
- SCADA Development
- Industrial Networking
- Instrumentation
- Automation Systems

Production Supervisor

Industrial Projects International Ltd., Abeokuta, Nigeria (Detergent Plant)

April 2022 – November 2024

- Managed production operations and process monitoring.
- Improved operational efficiency through data-driven decision making.
- Coordinated maintenance and production teams.

TECHNICAL SKILLS

- Programming: Python, SQL, C/C++, MATLAB
- Industrial Automation: PLC, HMI, SCADA, WinCC, TIA Portal
- Process Engineering: Aspen HYSYS, Aspen Plus, Design Expert
- Data Science: Pandas, NumPy, Scikit-Learn, Power BI, Excel

- Engineering Tools: AutoCAD, Process Simulation, Statistical Analysis

CERTIFICATIONS

- Data Science and AI – TedPrime Hub, Abeokuta, Nigeria
- Data Analysis & Visualization – Welup Digital Academy, Lagos, Nigeria
- HSE Levels 1, 2 and 3 & First Aid – Fest-Michelle Consultancy Services, Sango-Ota, Nigeria
- Python for Everybody – Coursera, Online

RESEARCH GOAL

To develop intelligent industrial systems that integrate machine learning, process control, industrial automation, and digital technologies for fault diagnosis, predictive maintenance, and smart manufacturing applications.

REFERENCE(S)

To be submit on request